

# Spatial ML - class 9 - Causal ML

Total points 100/100 ?

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**Which question is causal rather than descriptive or predictive?**

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- ☐ Which block groups have the highest building permit density?
- ☒ Did the 606 intervention increase building permit density compared with what would have happened otherwise?
- ☐ Which covariates are most correlated with building permit density?
- ☐ Which block groups are most likely to have high permit density next year?

**In the potential outcomes framework, what is the fundamental problem of causal inference?**

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- ☐ We do not know which model is best.
- ☒ We cannot observe both treated and untreated outcomes for the same unit.
- ☐ We do not know the sample size in advance.
- ☐ We cannot estimate correlations.



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**Which variable should usually be adjusted for when estimating a total treatment effect?**

- ☐ A variable that is only strongly correlated with the outcome after treatment
- ☒ A pre-treatment confounder affecting both treatment and outcome
- ☐ A post-treatment mediator
- ☐ A collider created by treatment and outcome

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**Why is a single Average Treatment Effect often not enough in applied spatial policy problems?**

- ☐ Because spatial data cannot be summarized numerically
- ☒ Because treatment effects may differ across places and populations
- ☐ Because causal forests do not estimate averages
- ☐ Because averages are never statistically valid



**What does SUTVA require?** \*

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- ☐ The outcome must be measured before treatment
- ☐ All covariates must be normally distributed
- ☒ No hidden versions of treatment and no interference between units
- ☐ Treatment must always be randomized

**Why is SUTVA especially difficult in spatial settings?** \*

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- ☐ Spatial data always have missing values
- ☒ Nearby units may be affected by the same intervention through spillovers
- ☐ Maps cannot show treatment clearly
- ☐ Spatial models cannot estimate treatment effects



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**In the causal forest practical, why do untreated observations also receive predicted CATE values?**

- ☐ Because the model incorrectly treats all units as treated
- ☐ Because CATE is just another name for the observed outcome
- ☐ Because untreated units are removed before modelling
- ☒ Because CATE is a model-based estimate of what the treatment effect would be for each unit

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**In the overlap diagnostic, the spatial propensity model produced much more extreme probabilities than the baseline model. What does that suggest?**

- ☐ The spatial model is misspecified
- ☐ There is no longer any heterogeneity
- ☐ The treatment effect is definitely zero
- ☒ Treated and untreated units may be less comparable once space is included



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**In the comparison plot of baseline CATE vs spatial CATE, most points were below the 45-degree line. What is the best interpretation?**

- ☐ The spatial model predicts larger effects for most units
- ☐ The baseline and spatial models are identical
- ☒ The spatial model predicts smaller treatment effects for most units
- ☐ Only treated units receive predicted effects

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**In the class example, why did adding spatial features reduce the estimated treatment effect?**

- ☐ Because adding more variables always lowers treatment effects
- ☐ Because causal forests cannot handle coordinates
- ☒ Because some of what looked like treatment effect in the non-spatial model may actually have reflected spatial structure or spatial sorting
- ☐ Because spatial features automatically remove all bias



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